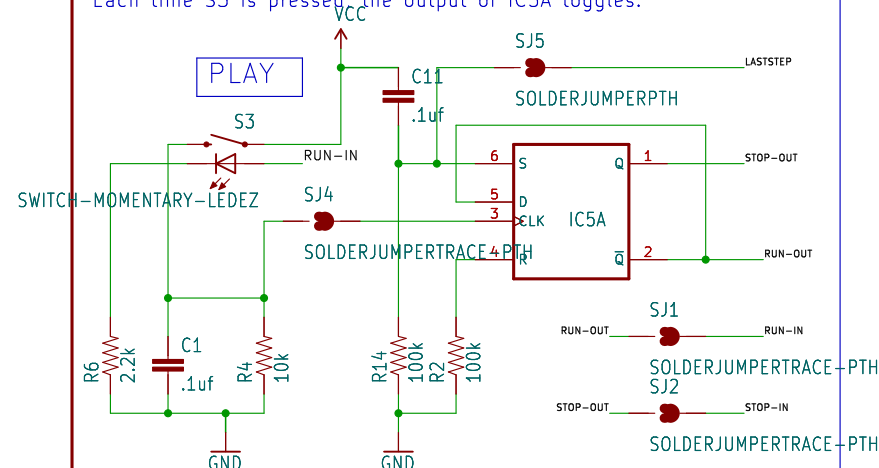
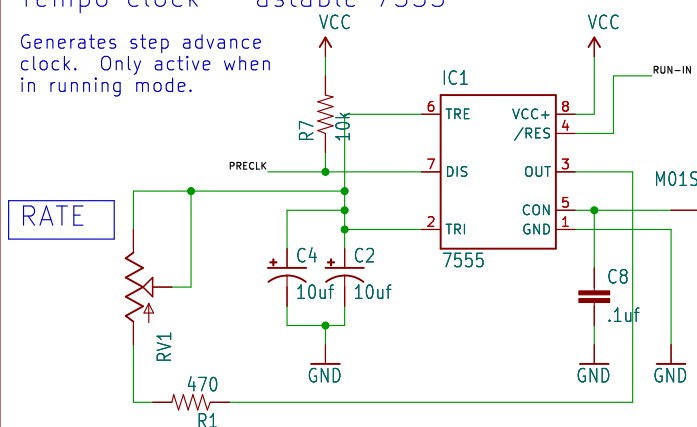


Start pushbutton and toggle flip-flop
Each time S3 is pressed, the output of IC5A toggles.

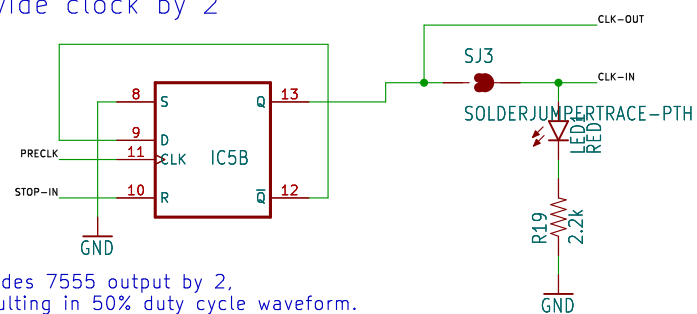


Tempo clock – astable 7555

Generates step advance clock. Only active when in running mode.



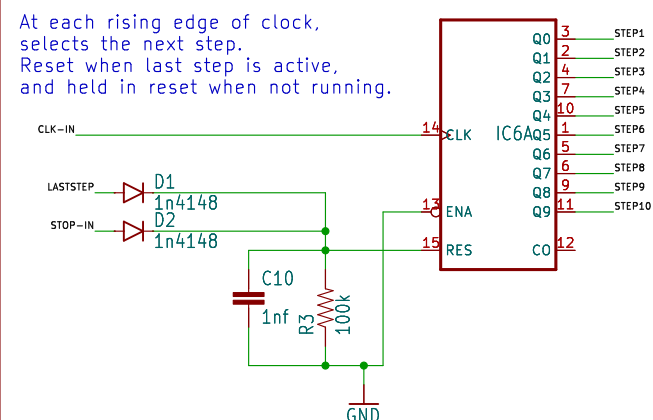
Divide clock by 2



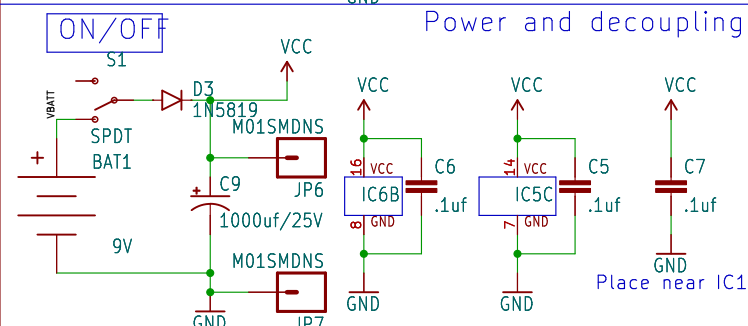
Divides 7555 output by 2, resulting in 50% duty cycle waveform.

1 of 10 Counter/Divider

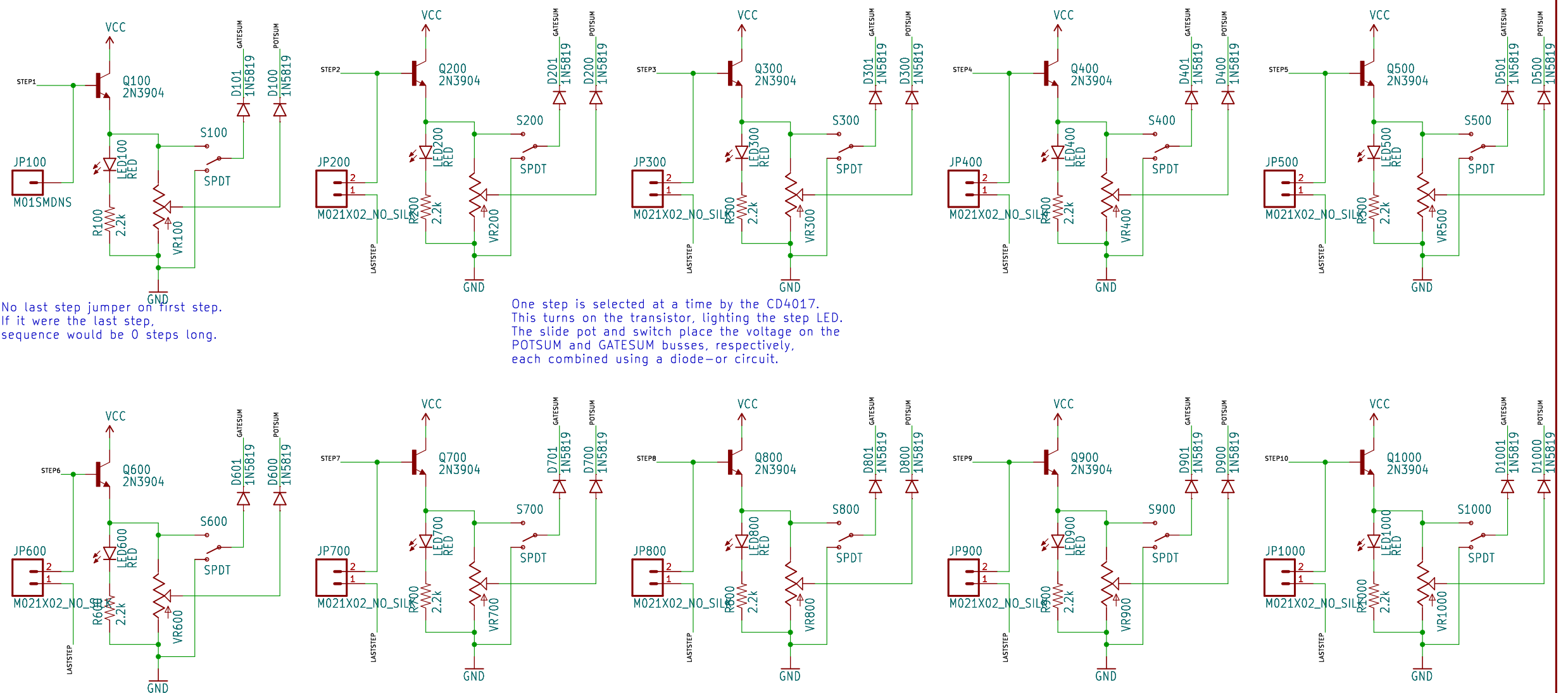
At each rising edge of clock, selects the next step. Reset when last step is active, and held in reset when not running.



Power and decoupling



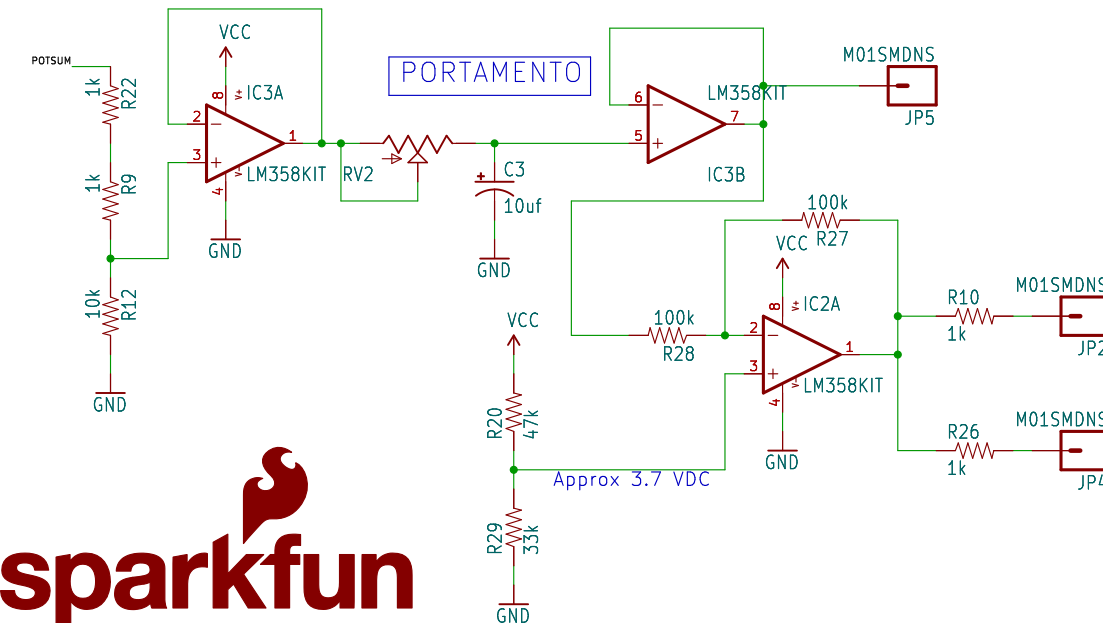
10 steps each with LED, control voltage slider, gate switch and last step jumper



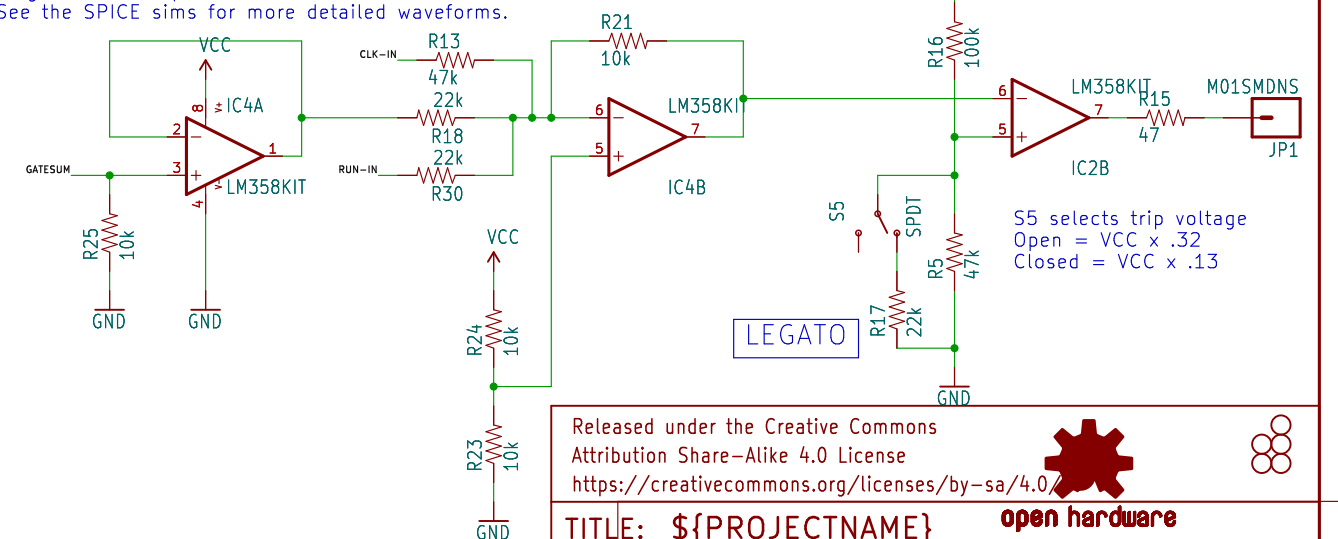
No last step jumper on first step. If it were the last step, sequence would be 0 steps long.

One step is selected at a time by the CD4017. This turns on the transistor, lighting the step LED. The slide pot and switch place the voltage on the POTSUM and GATESUM busses, respectively, each combined using a diode-or circuit.

Control Voltage summing & glide circuit
IC3G1 scales and buffers the voltage from the slider wipers – scaled to a range that doesn't clip the opamp by getting too close to the 9V rail.
RV2 and C3 form an adjustable lowpass on the resulting voltage, buffered by IC3G2.
IC2G1 inverts, shifts and scales that to a range suitable to drive 555's.
A 555's CV range is centered between the rails, and has an inverse voltage-to-frequency response.



Gate output processing
IC4G1 buffers the gate switch output.
IC4G2 adds scaled versions of the 50% step clock and running signal. The result is a stair-step waveform of clock superimposed on gate.
IC2G2 is a comparator that trips when the stair-step waveform crosses a threshold. The threshold can be set for long or short steps.
See the SPICE sims for more detailed waveforms.



S5 selects trip voltage
Open = VCC x .32
Closed = VCC x .13

